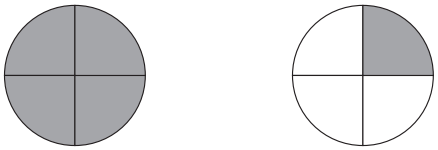
Rational Number Project

Initial Fraction Ideas Lesson 14: Overview	Materials
Students name fractions greater than 1 but less than 2 with fraction circles. Students name fractions using both mixed numbers and improper fractions.	• Fraction Circles for students and teacher • Transparency 1 • Student Pages A and B

Teaching Actions	Comments
Warm Up You and your friends ate one whole pizza and 1-third of another one. Draw a picture of this amount of pizza. How much pizza is this?	Modeling fractions greater than one using fraction circles, is easier than with chips, so we concentrate on developing the concept of changing improper fractions and vice versa with fraction circles.
Large Group Introduction 1. Ask students to use their fraction circles with the black circle as the unit to show $\frac{2}{2}$, $\frac{4}{4}$, $\frac{5}{5}$ and $\frac{12}{12}$. In each example, ask for another name for the amount shown (1 whole or just 1).	The warm up sets the stage for the lesson dealing with fractions greater than one. Discuss together how students named the fraction amount. Explain that in this lesson they will find ways to name fractions > 1.
2. Have students show $\frac{6}{8}$ using the whole circle as the unit. Ask if $\frac{6}{8}$ is greater or less than 1 whole or 1?	
3. Present this story and ask students to model it with their circles. Again, use whole circles as the unit. *Last night Margo ate $\frac{3}{4}$ of a large pizza. (Show that with circles). In the morning she ate some leftover pizza that equaled $\frac{2}{4}$ of a pizza.*	Accept both names: 1 and $\frac{1}{4}$ or $\frac{5}{4}$. Do not rush any rules about changing improper fractions to mixed fractions. Our goal is for students to change from one notation to another using circles and then just with mental images of circles. No paper/pencil rules.
4. Ask students to try and show the extra $\frac{2}{4}$. They realize that they do not have enough pieces. Have them work with a partner and use 2 sets of fractions circles to model the story.	
5. Continue with the story: How much pizza did	

Teaching Actions

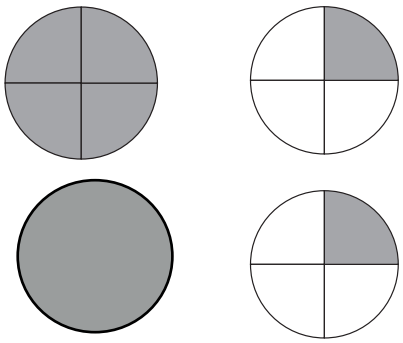
Margo eat altogether?



Questions to lead discussion for naming amount of pizza:

- Did Margo eat more than 1 whole pizza? How do you know?
- Let's count how many fourths she ate: $\frac{1}{4}$, $\frac{2}{4}$, $\frac{3}{4}$, $\frac{4}{4}$, $\frac{5}{4}$.
- From the picture I see that $\frac{5}{4}$ equals $\frac{4}{4}$ and _____.
- What's another name for $\frac{4}{4}$?
- What's another way of describing the amount of pizza Margo ate?

6. Draw a picture of what you did and restate how $\frac{5}{4} = 1\frac{1}{4}$.



This shows $\frac{5}{4}$; point to $\frac{4}{4}$ and $\frac{1}{4}$ more.

This shows $\frac{5}{4}$; 1 whole and $\frac{1}{4}$ of another circle.

6. Ask students to use their fraction circles to show these amounts. If the fraction >1, have the students name the amount in two ways.

$\frac{4}{3}$

$\frac{3}{4}$

Comments

Students may try to name the picture of two circles with 5 parts shaded as $\frac{5}{8}$. If they do this, ask: What is the unit?

Since the unit is one pizza, each part is named by comparing it to one pizza or one circle.

Teaching Actions

$\frac{3}{2}$

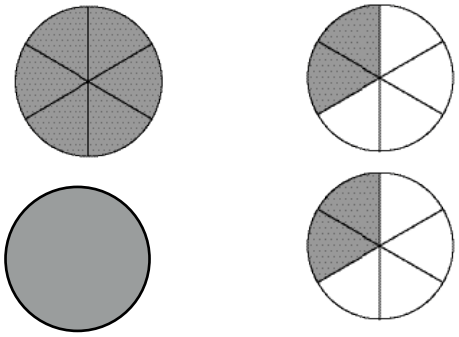
$1\frac{2}{3}$

$\frac{5}{6}$

7. As students explain their models, ask if the amount is greater or less than one. Try to get them to verbalize concrete actions that show when a fraction is greater than one.

8. Select students to draw pictures for a few examples showing two ways to name the fraction.

Ex: $\frac{8}{6}$



Small Group/Partner Work

10. Assign Student Pages A, B, C.

Wrap Up

11. Ask students to imagine each fraction noted below and from their mental image name the amount in another way.

$\frac{4}{3}$ $\frac{3}{2}$ $\frac{3}{3}$

Comments

We want students to verbalize that, for example, $\frac{8}{6}$ is greater than 1 because they need more than 1 whole circle to model it. $\frac{4}{5}$ is less than 1 because they needed only 1 unit to model it.

Translations

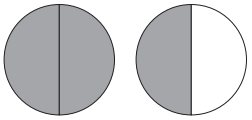
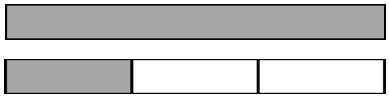
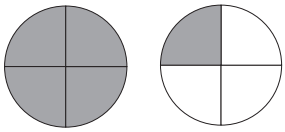
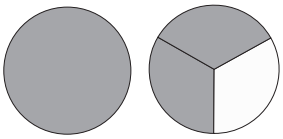
- Real world to manipulative to verbal
- Written symbols to manipulative to written symbols
- Written symbols to picture to verbal to written symbols

You and your friends ate one whole pizza and 1-third of another one.

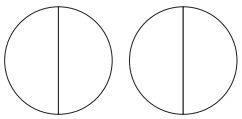
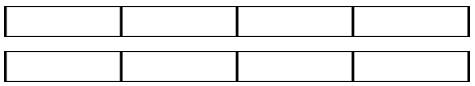
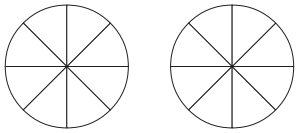
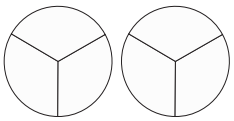
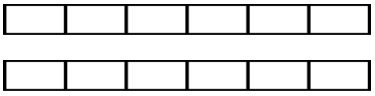
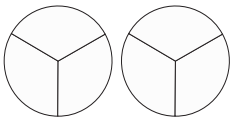
Draw a picture of this amount of pizza.

How much pizza is this?

Write two fraction names for each picture.

Improper Fraction		Mixed Number
		
		
		
		
		
		

Shade in the pictures to show each fraction. Write another name for each amount

Mixed Number	Shade In	Improper Fraction
$1\frac{1}{2}$		
$1\frac{3}{4}$		
		$\frac{11}{8}$
		$\frac{5}{3}$
$1\frac{4}{6}$		
		$\frac{4}{3}$